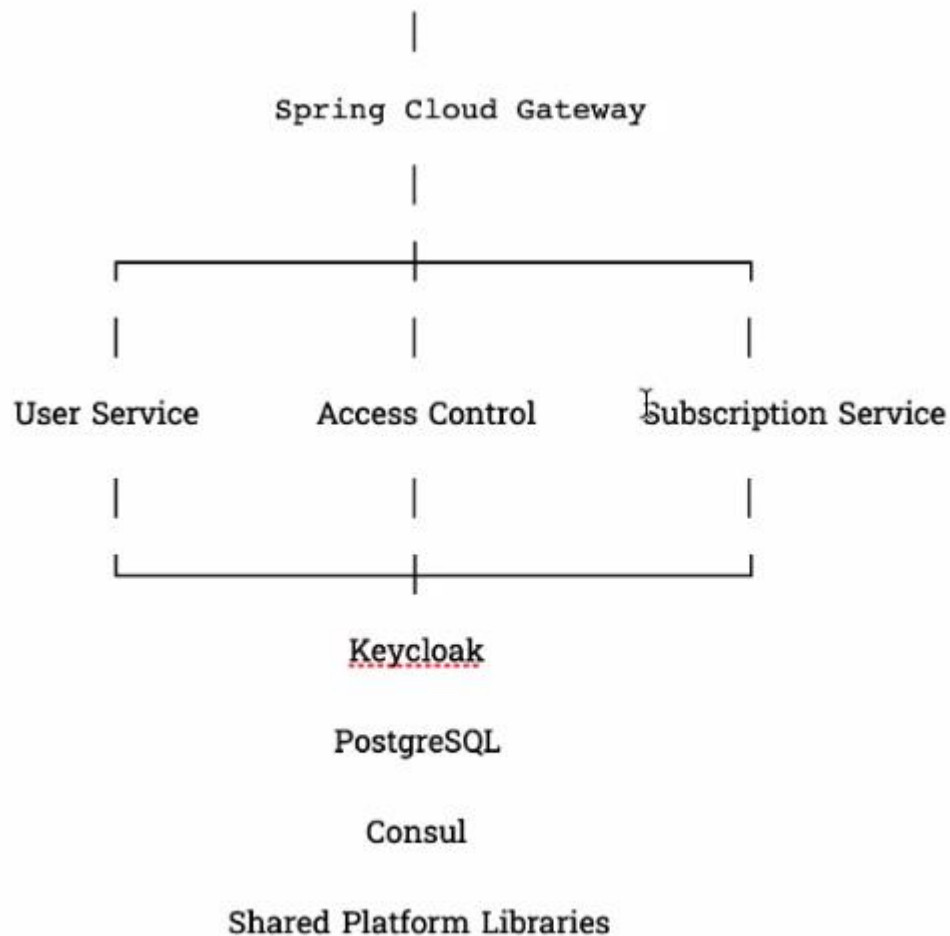


Enterprise Backend Development with Spring Boot 4

Build a Real Subscription Platform

What students will build



- Security: We use centralized using keycloak (reduce cross cutting concern)
- User Service: សម្រាប់ធ្វើការ Register
- Access Control: user can access to which service
- Subscription Service: Saas; example house owner and end user សម្រាប់ធ្វើការ control which package they should have
- Upgrade business level: កន្លែងកំសាន្ត KTV, កន្លែងញ៉ាំអី (អាចអោយអតិថិជនធ្វើការស្វែងរក កន្លែងដែល ញ៉ាំអី នឹងមានអ្នកចូលមើលច្រើន)
- Spring cloud Gateway ពេលចូលមកអោយត្រូវវាចូលតាម Gateway របស់យើង ហើយ
- Service registration for each service: we use Consul (like the Urica service)
- Shared Platform Libraries: share code

What students will learn

Module 1: Enterprise Project Setup

- Spring Boot 4
- Java 25
- Maven Multi Module
- Consul
- PostgreSQL
- Docker
- Keycloak

Module 2: Platform Libraries

Students build

platform-common

platform-security

platform-web

platform-logging

platform-exception

platform-openapi

platform-access-control-client

- Module 2: Platform Libraries

Module 3: Authentication

Students learn

- OAuth2
- JWT
- Access Token
- Refresh Token
- Keycloak Realm
- Client
- User
- Resource Server

Module 4: Gateway

Students build

Gateway -> Validate Token -> Forward Token -> Subscription Service

Learn

- Security
- Routing
- Swagger
- Global Exception
- Logging

Module 5: Access Control

Students build

Permission -> Role -> User -> Authorization -> API Protection

Module 6: Subscription Service

This is the main project.

Example entities: Plan, Subscription, Invoice, Payment, Feature, Organization, Member, Usage.

Module 7: Spring Security

Students learn

I

hasRole()

hasAuthority()

Permission Evaluator

Method Security

Custom Authorization

Module 8: Forward JWT

One of the hardest topics.

Gateway -> Subscription -> User Service -> Access Control -> Token Forwarding

Module 9: Logging

SLF4J

MDC

Trace ID

Request ID

Error Log

Audit Log

Production Logging

Module 10: Swagger

Gateway Swagger

Service Swagger

OpenAPI

Aggregation

Exactly what companies use.

Module 11: Design Patterns Usage

- Strategy
- Factory
- Builder
- Facade
- Template Method

Module 12: Best Practices

DTO

Mapper

Facade

Service

Repository

Validation

Exception

Package Structure

Naming

Transactions

Technologies

Technology	Students Learn
Java 25	Modern Java
Spring Boot 4	Enterprise Backend
Spring Security	API Security
Keycloak	Authentication
OAuth2	JWT
PostgreSQL	Database
Liquibase	Database Migration
MapStruct	Mapping
Gateway	API Gateway
Consul	Service Discovery
OpenAPI	Swagger
Docker	Deployment
MDC Logging	Enterprise Logging
Design Patterns	Clean Architecture